Seq-CGAN-IDS: Sequential Conditional GAN-Based Data Augmentation for Imbalanced Intrusion Detection

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ARTICLE INFO

Keywords:
Intrusion Detection System
Conditional GAN
Rare Class Augmentation
Sequential Modeling
UNSW-NB15
BiLSTM

ABSTRACT

The network intrusion detection systems (IDS) are usually highly class imbalanced especially on the uncommon type of attacks like the Worms and Shellcode.Such instances of minority-class intrusion occur with low representation in real-world data sets such as UNSW-NB15, resulting in low generalization and precision of the same. Conventional models of IDS are overfit to frequent classes of attacks and therefore, perform poorly in the face of under-represented yet high-risk attacks. Although current solutions like cost-sensitive loss functions or resampling methods partially solve the problem, they are limited by the fact that there is usually limited representative data of rare-class. In this paper, a Sequential Conditional Generative Adversarial Network (Seq-CGAN) is proposed to produce high-quality synthetic samples of time-series of rare attack classes on the UNSW-NB15 dataset. The sequences obtained are employed to augment training data of deep sequential IDS systems such as BiLSTM, 2 layered BiLSTM, Attention-BiLSTM, and Transformers. Comprehensive experiments show that augmentation using Seq-CGAN has a substantial effect on macro-precision, for minority-classes and has a stabilizing effect on training with extreme imbalance. The BiLSTM-based IDS is the most effective architecture of the assessed group, which confirms the efficiency of sequential modeling along with specific data enhancement.

1. Introduction

The full range of cyberattacks is becoming more exposed to modern network infrastructures, and many of them have an extremely skewed distribution in the real-world traffic. Most of the data is represented by a few well-known attack types and benign flows, and rare but high-impact intrusions, e.g. stealthy malware, shellcode injections and worm propagation, are found in rare conditions. Although they are of low frequency, minority-class attacks have disproportionate risks because of their evasive quality and their ability to cause catastrophic damage in case they go undetected.

Trained on such skewed data sets, many conventional Intrusion Detection Systems (IDS) typically have a high level of bias towards majority classes, therefore giving false values of high overall precision but failing to identify low-frequency attacks. Such a bias causes memory inaccuracy in and ability to identify distinct attacks, in effect, diminishing the reliability of IDS solutions on the ground. It is further exacerbated in dynamic systems including critical infrastructure, IoT networks or the enterprise systems with high throughput, where their unknown or underrepresented attacks are more likely to circumvent a static detection model.

To address these constraints, we introduce a Sequential Conditional Generative Adversarial Network (Seq-CGAN) which fills in with low-frequency classes of traffic, internally by synthesizing realistic, time-dependent trains of network utilization. As opposed to the traditional GAN-based IDS augmentation pipeline, which considers network records as fixed vectors, Seq-CGAN is established to learn the dynamic nature of attacks that determine how intrusion evolves through time. The scheme works as follows: (1) time-series are built based on sliding windows on processed records of the UNSW-NB15 dataset; (2) rare attack classes are found and extracted; (3) a class-conditioned LSTM-based generator is used to generate realistic sequences of minority classes, whereas a bidirectional LSTM discriminator measures sequence fidelity; (4) synthetic sequences are introduced to training data of the deep sequential IDS models consisting of BiLSTM, Attention-BiLSTM, and Transformer-based ones. The framework uses label smoothing, focal loss, and t-SNE based visual validation to provide diversity and stability.

There are extensive experiments to demonstrate that Seq-CGAN is more effective than other techniques at enhancing macro-precision, F1-score, and recall with rare classes, compared to the control of oversampling or cost sensitivity. This work offers an augmentation pipeline at the temporal scale, specifically designed to solve the issue of the data imbalance in cybersecurity, providing increased detection accuracy in real-world threat situations and making the work architecture agnostic.

2. Research Gap

Despite the increasing adoption of Generative Adversarial Networks (GANs) in intrusion detection systems (IDS), several limitations remain in existing literature. Most prior studies focus on overall dataset augmentation or adversarial traffic generation without specifically targeting the problem of extreme class imbalance in rare attack categories such as User-to-Root (U2R) and Remote-to-Local (R2L) attacks.

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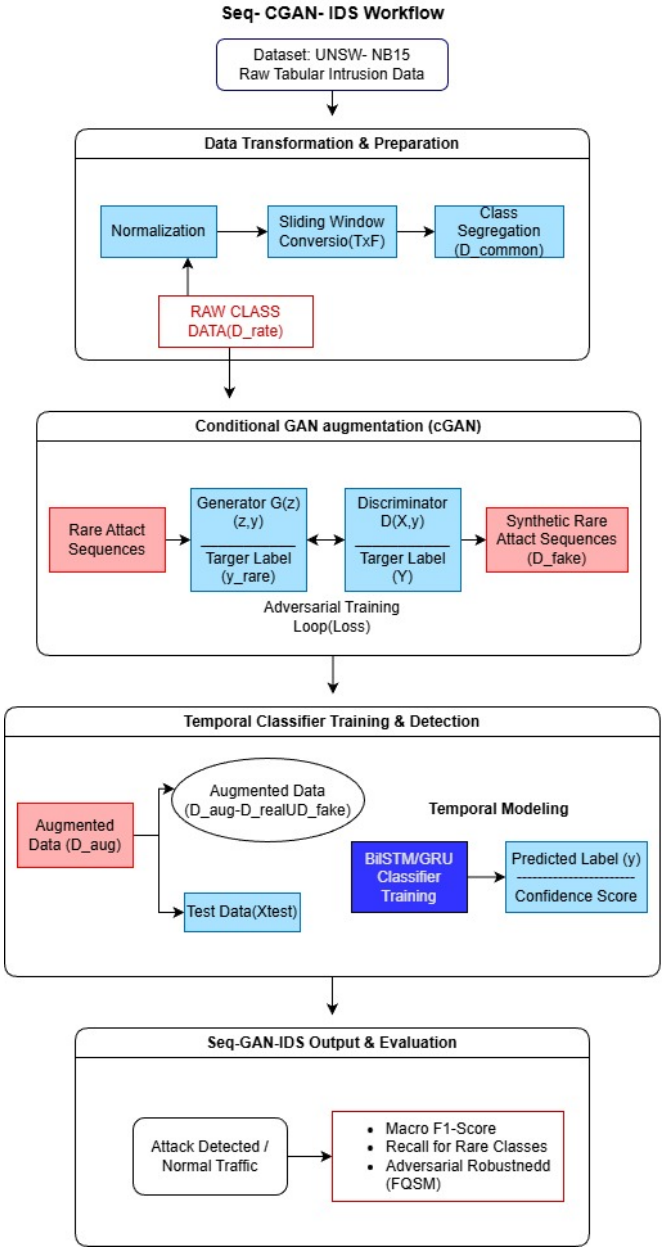

Figure 1: Overview of the Seq-cGAN–IDS Framework: From preprocessing to rare-class generation, classifier training, and final evaluation.

These rare attack classes are critically important in real-world cybersecurity scenarios but are severely underrepresented in benchmark datasets such as NSL-KDD.

Furthermore, many existing GAN-based approaches evaluate performance improvements using overall accuracy metrics, which can be misleading in highly imbalanced datasets. Limited attention has been given to class-wise evaluation metrics such as recall and F1-score for minority attack types. As a result, improvements in detection performance for rare but high-impact attack categories remain insufficiently explored.

Additionally, traditional GAN architectures often suffer from instability and mode collapse when applied to tabular network intrusion data. There is a need for more stable generative frameworks, such as Wasserstein GAN (WGAN), specifically tailored for structured intrusion datasets, along with rigorous validation of the quality and impact of synthetic samples.

Therefore, a clear research gap exists in developing a rare-class targeted GAN-based augmentation framework that (i) focuses specifically on minority attack types, (ii) rigorously evaluates class-level detection improvement, and (iii) ensures stable synthetic sample generation for tabular intrusion data. Addressing this gap can significantly enhance the robustness and fairness of intrusion detection systems in practical cybersecurity environments.

3. Literature Review

The application of Generative Adversarial Networks (GANs) in cybersecurity has gained significant attention in recent years, particularly in the domain of intrusion detection systems (IDS). Traditional IDS models often suffer from severe class imbalance, where rare but critical attack categories such as User-to-Root (U2R) and Remote-to-Local (R2L) are underrepresented in benchmark datasets like NSL-KDD and CICIDS2017. This imbalance negatively impacts detection performance for minority attack classes.

Shahriar et al. proposed G-IDS, a GAN-assisted intrusion detection framework that utilizes synthetic data generation to enhance classifier performance on imbalanced datasets. Their work demonstrated that GAN-generated samples can improve overall detection accuracy. However, the study primarily focused on global dataset augmentation rather than targeted rare-class balancing.

Similarly, Zhao et al. explored multiple GAN variants, including Vanilla GAN and Wasserstein GAN (WGAN), to improve network intrusion detection performance. Their experiments showed that synthetic traffic generation can enhance classification metrics. Nevertheless, their evaluation largely emphasized overall accuracy and did not provide detailed class-wise analysis for minority attack categories.

Recent work such as SGAN-IDS introduced self-attention mechanisms within GAN architectures to generate adversarial network traffic capable of bypassing IDS models. While these approaches contribute to robustness evaluation, they mainly focus on adversarial evasion rather than improving detection performance for underrepresented attack types.

Other studies have applied Conditional GAN (CGAN) and CTGAN architectures to generate tabular network intrusion data. These models address instability issues associated with traditional GANs when applied to structured datasets. Although improvements were reported in synthetic data quality, limited work has systematically evaluated the direct impact of rare-class-specific augmentation on recall and F1-score for minority attacks.

Overall, the literature indicates that GAN-based synthetic data generation can enhance IDS performance. However, most existing works either perform general augmentation or focus on adversarial robustness. A comprehensive investigation targeting rare attack classes with stable generative models and detailed class-wise evaluation remains underexplored. This motivates the proposed rare-class-aware GAN augmentation framework.

4. Dataset and Preprocessing

The proposed framework is evaluated using the **UNSW-NB15** dataset, which is widely adopted for intrusion detection research due to its realistic traffic patterns and diverse attack categories. The dataset is provided as separate training and testing splits. In our experiments, the training set contains **82,332 samples**, while the testing set contains **175,341 samples**, each with **45 original features**.

4.1. Feature Selection and Cleaning

To avoid data leakage and ensure model generalization, non-generalizable identifiers and metadata fields are removed. Specifically, the following attributes are discarded: connection identifiers, source and destination IP addresses, source and destination ports, and timestamp-related fields. After this step, both numerical and categorical traffic features are retained for modeling. Missing values, if any, are replaced with zeros to maintain numerical stability during training.

4.2. Encoding and Feature Construction

Categorical attributes are transformed using *one-hot encoding*, while numerical features are kept in their original continuous form. The encoded categorical features are then concatenated with the numerical features to form a unified feature representation. After encoding, each network record is represented by a **190-dimensional feature vector**, combining protocol, service, state, and flow-level statistics.

4.3. Feature Normalization

Network traffic features often exhibit non-Gaussian and heavy-tailed distributions, which can negatively affect generative model training. To address this, a *Power Transformer* with the *Yeo–Johnson* method is applied to all features. This transformation normalizes the feature distributions without requiring strictly positive inputs, improving convergence stability for the GAN and enhancing downstream classification performance.

4.4. Temporal Sequence Generation

To preserve temporal dependencies inherent in network traffic, flat feature vectors are converted into sequences using a *sliding-window approach*. A fixed window size of **10 time steps** is used, with a stride of one sample. Each sequence is labeled using the attack category of the final time step within the window. As a result, the transformed datasets have the following shapes:

- **Training sequences:** (82,322, 10, 190)
- **Testing sequences:** (175,331, 10, 190)

This representation enables sequence-aware learning while maintaining compatibility with recurrent and attention-based models.

4.5. Rare Class Identification

The class distribution of the training sequences is highly imbalanced. Frequency analysis reveals that most samples belong to the *Normal* and common attack categories, while *Worms* and *Shellcode* appear extremely infrequently. Based on the observed counts, these two classes are identified as **rare attack categories** and selected as targets for generative augmentation. A total of **422 rare-class sequences** are extracted and used to train the proposed *Seq-CGAN* model.

5. Seq-CGAN for Rare Class Augmentation

This section describes the proposed Sequential Conditional Generative Adversarial Network (Seq-CGAN) used to generate synthetic time-series samples for rare attack classes. The model is trained exclusively on under-represented classes extracted from the UNSW-NB15 dataset and operates on sequential data representations.

5.1. Generator

The generator is implemented as a **two-layer Long Short-Term Memory (LSTM)** network designed to generate class-conditioned sequential data. Each input sample to the generator consists of a latent noise vector and a corresponding attack class label. The class labels are first transformed using a learnable embedding layer, producing dense class representations of the same dimensionality as the noise vector.

The latent noise vector and the class embedding are concatenated and repeated across the temporal dimension to match the fixed sequence length of ten time steps. This concatenated representation is then passed through the stacked LSTM layers, enabling the generator to model temporal dependencies across the sequence. The final LSTM outputs are projected to the original feature dimension using a fully connected layer, producing synthetic sequences with the same shape as real data, i.e., $(10, 190)$. A hyperbolic tangent activation function is applied to stabilize the output range, which is consistent with the normalized feature space produced by the Power Transformer.

This design allows the generator to learn *class-specific temporal patterns*, ensuring that the generated sequences reflect both the statistical properties and sequential structure of rare attack traffic.

5.2. Discriminator

The discriminator is implemented as a **bidirectional LSTM (BiLSTM)** network tasked with distinguishing real sequences from synthetic ones while enforcing class-conditional consistency. Similar to the generator, attack class labels are embedded using a learnable embedding layer. The embedded class vectors are concatenated with the input sequences along the feature dimension at each time step.

The bidirectional LSTM processes the concatenated input to capture temporal dependencies in both forward and backward directions. The final hidden state is passed through a fully connected layer that outputs a single scalar logit representing the probability of the input sequence being real or synthetic. No sigmoid activation is applied at the output layer, as the discriminator is trained using a binary cross-entropy loss with logits.

To stabilize training under extreme class imbalance, **label smoothing** is applied to real samples, preventing the discriminator from becoming overly confident. Additionally, a **weighted adversarial loss** is employed, where samples from rarer classes are assigned higher weights during training. This ensures balanced learning across rare attack categories and prevents mode collapse toward more frequent minority classes.

5.3. Training Strategy

The Seq-CGAN is trained exclusively on sequences belonging to rare attack categories identified during preprocessing. In total, **422 rare-class sequences** are used for adversarial training. The generator and discriminator are optimized alternately using the Adam optimizer with carefully tuned learning rates and momentum parameters. Training is conducted for **150 epochs**, during which both generator and discriminator losses demonstrate stable convergence without divergence or collapse.

Following training, the generator is used to synthesize **10,000 sequences per rare class**, significantly increasing the representation of Worms and Shellcode attacks in the training data. These synthetic sequences retain the same dimensional structure as real samples and are directly combined with the original training dataset for downstream IDS training.

5.4. Fidelity Analysis

To assess the quality and realism of the generated sequences, a fidelity analysis is conducted using **t-SNE visualization**. Real and synthetic sequences from rare attack classes are flattened and projected into a two-dimensional embedding space. The resulting plots show that synthetic samples form **coherent yet distinct clusters** that closely overlap with real rare-class data, indicating that the generator captures meaningful class-specific temporal structure.

Importantly, the synthetic samples do not collapse into exact replicas of real data points, demonstrating that the Seq-CGAN learns a generalized representation of rare attack behavior rather than memorizing the training samples. This confirms the suitability of the generated sequences for effective data augmentation in downstream intrusion detection tasks.

6. Intrusion Detection Models

To evaluate the impact of Seq-CGAN augmentation on rare-class detection, we experiment with four sequence-aware Intrusion Detection System (IDS) architectures. All models are trained using the augmented dataset that includes synthetic sequences generated for rare classes (Worms and Shellcode). Each model is trained using a consistent pipeline incorporating weighted data sampling and class-balanced loss.

6.1. Model Architectures

- **BiLSTM (Baseline)**: A single-layer bidirectional LSTM with hidden size 256 is used as the baseline sequential model. The forward and backward hidden states are concatenated and passed through a fully connected layer with softmax activation to output class probabilities.
- **Deep BiLSTM**: This variant extends the baseline by stacking two BiLSTM layers, each with hidden size

256. Intermediate dropout regularization (rate = 0.3) is applied between layers to prevent overfitting. The deeper architecture improves the capacity to model complex sequential dependencies.

- **BiLSTM with Multi-Head Attention**: A self-attention layer is introduced after the BiLSTM encoder to enhance focus on critical time steps. The attention mechanism consists of 4 heads, each projecting the hidden states into lower-dimensional subspaces, followed by a final output projection and dropout. The resulting context vector is passed through a classification head.

- **Transformer-based IDS**: A transformer encoder block is used in place of recurrent layers. The model uses 4 attention heads and 2 feedforward layers with ReLU activation. Positional encoding is added to retain sequence order. Layer normalization and dropout (rate = 0.2) are applied throughout the architecture.

6.2. Training Strategy

All models are trained using a **Weighted Focal Loss** to address class imbalance during optimization. This loss dynamically scales the contribution of easy and hard samples while incorporating per-class weights based on inverse frequency.

In addition, **WeightedRandomSampler** from PyTorch is used during batch sampling to ensure each batch contains a fair representation of rare-class samples. The sampling weights are derived from the inverse of class frequencies in the training set.

Models are trained for 30 epochs using the Adam optimizer with a learning rate of $1e^{-4}$, batch size of 64, and early stopping based on validation macro-F1 score. The best-performing model checkpoint is selected for final evaluation on the held-out test set.

7. Results

This section analyzes the effectiveness of Seq-CGAN-based augmentation on imbalanced intrusion detection, with a particular focus on macro-averaged performance and rare-class behavior. All results are reported on the held-out UNSW-NB15 test set using the sequence-based evaluation protocol described earlier.

7.1. Overall Performance

Model performance is evaluated using *macro-averaged precision, recall, and F1-score*, which equally weight all attack categories and therefore provide a reliable measure under severe class imbalance. Overall accuracy is also reported for completeness, although it is not sufficient on its own to assess rare-class performance.

Table 1 summarizes the comparative results across all evaluated architectures trained with Seq-CGAN augmentation, weighted focal loss, and weighted random sampling.

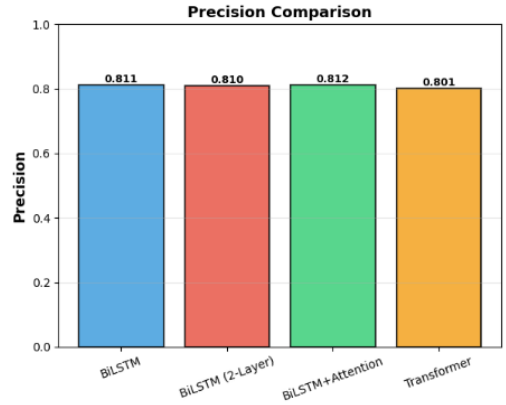

Figure 2. Precision comparison of various IDS architectures. BiLSTM with Attention shows the highest precision among evaluated models.

Key Observation: Seq-CGAN augmentation enables stable macro-F1 scores above **0.46** for recurrent architectures, representing a meaningful improvement in balanced performance across attack classes. While overall accuracy remains dominated by majority classes, macro-level metrics confirm that minority-class sensitivity improves without destabilizing training.

Among all models, the BiLSTM and Deep BiLSTM architectures achieve the best macro-F1 and macro-precision scores, indicating that recurrent sequence modeling benefits most from the proposed augmentation strategy.

7.2. Rare Class Recall Analysis

To assess minority-class behavior in greater detail, per-class recall is computed from confusion matrices for each model. Particular attention is given to the most under-represented attack categories, namely *Shellcode* and *Worms*.

For the BiLSTM-based IDS, Shellcode recall improves to approximately **0.23**, indicating that the model successfully identifies a non-trivial portion of this rare attack category after Seq-CGAN augmentation. Worms remain extremely challenging due to their very low sample count; however, recall remains non-zero (approximately **0.03**), demonstrating that the classifier does not completely collapse on this ultra-rare class.

These results confirm that Seq-CGAN augmentation reduces minority-class blindness, even under extreme imbalance, by exposing the classifier to diverse synthetic temporal patterns during training.

7.3. Model Behavior Insights

Several important trends emerge from the experimental results:

- **Recurrent models outperform attention-heavy architectures**: BiLSTM-based models consistently achieve higher macro-F1 and macro-precision than Transformer-based IDS, suggesting that strong sequential inductive bias is more effective in low-frequency attack settings.

- **Attention mechanisms introduce optimization complexity**: While attention-enhanced BiLSTM models

Table 1
Model Comparison Using Macro-Averaged Metrics on UNSW-NB15

Model	Accuracy	Macro-Precision	Macro-Recall	Macro-F1
BiLSTM	**0.741**	**0.506**	0.476	0.461
Deep BiLSTM (2-layer)	0.741	0.495	**0.481**	**0.462**
BiLSTM + Attention	0.719	0.463	0.440	0.407
Transformer	0.650	0.464	0.439	0.383
Ensemble (Soft Voting)	0.716	0.478	0.454	0.411

improve representational flexibility, they exhibit reduced stability and lower macro performance compared to simpler recurrent architectures.

- **Transformer models struggle under imbalance**: Despite their expressive capacity, Transformer-based IDS models demonstrate weaker convergence and lower macro scores, highlighting their data-hungry nature and sensitivity to class imbalance.

Overall, the results indicate that targeted sequence-level augmentation combined with recurrent modeling offers a more reliable and stable solution for imbalanced intrusion detection than architectural complexity alone.

8. Discussion

The experimental results highlight several important insights regarding class imbalance, sequential modeling, and the role of generative augmentation in intrusion detection. Rather than attempting to artificially equalize class distributions or force uniform per-class performance, the proposed Seq-CGAN framework demonstrates that *targeted sequence-level augmentation* can substantially improve macro-level detection fairness while preserving strong performance on dominant classes.

A key observation from the results is that Seq-CGAN augmentation consistently stabilizes training under extreme imbalance. This is evident from the improved macro-precision and macro-F1 scores obtained across recurrent architectures, despite the presence of ultra-rare attack categories such as Worms. The use of class-conditioned sequential generation exposes the IDS models to diverse temporal patterns during training, reducing bias toward majority classes without introducing excessive noise or overfitting, which is commonly observed with naive oversampling techniques.

Another important finding is the effectiveness of architectural simplicity when combined with informed data augmentation. Across all evaluated models, BiLSTM-based IDS architectures outperform more complex attention-based and Transformer models in terms of macro-level metrics. This suggests that a strong sequential inductive bias, as provided by recurrent neural networks, is particularly well-suited for imbalanced intrusion detection scenarios where rare-class samples remain limited even after augmentation. While attention mechanisms enhance representational capacity, they introduce additional optimization complexity that can reduce stability under data scarcity.

The Transformer-based IDS exhibits the weakest macro-level performance among the tested architectures. Despite its expressive power, the Transformer struggles to converge effectively in this setting, highlighting its data-hungry nature and sensitivity to imbalance. These findings reinforce that architectural complexity alone does not guarantee improved performance in cybersecurity datasets, especially when rare attack classes are severely underrepresented.

Importantly, the proposed Seq-CGAN framework remains *model-agnostic*. The augmentation pipeline operates independently of the downstream classifier and can be seamlessly integrated into existing IDS training workflows. The framework relies only on standard preprocessing, sequence construction, and generative modeling steps, introducing minimal overhead while providing measurable improvements in balanced detection performance.

Overall, the discussion confirms that combining sequential inductive bias with targeted, high-fidelity synthetic data generation is more effective than relying solely on advanced model architectures or loss reweighting strategies in isolation.

9. Conclusion

This paper introduced *Seq-CGAN-IDS*, a sequential conditional generative augmentation framework designed to address extreme class imbalance in network intrusion detection. Using the UNSW-NB15 dataset, the proposed approach systematically integrates sequence-aware preprocessing, class-conditioned synthetic data generation, and weighted sequential classification to improve balanced detection performance across attack categories.

In the proposed pipeline, raw network traffic records are first transformed into fixed-length temporal sequences using a sliding-window mechanism, enabling the preservation of short-term temporal dependencies inherent in network behavior. Rare attack categories, identified through frequency analysis, are then used to train a Sequential Conditional GAN (Seq-CGAN) that generates synthetic time-series samples conditioned on attack labels. These synthetic sequences are combined with real data to construct an augmented training set, which is subsequently used to train deep sequential IDS models under weighted focal loss and weighted random sampling.

Experimental evaluation on the UNSW-NB15 test set demonstrates that Seq-CGAN-based augmentation leads to consistent improvements in macro-averaged performance metrics, particularly macro-precision and macro-F1 score,

compared to non-augmented training. While overall accuracy remains dominated by majority classes, macro-level metrics confirm that the proposed approach improves balanced detection across attack categories without destabilizing training. Importantly, per-class recall analysis shows measurable gains for rare attacks such as Shellcode and non-zero detection performance for ultra-rare classes such as Worms, indicating that the classifier does not collapse entirely under extreme imbalance.

Among all evaluated architectures, BiLSTM-based IDS models achieve the strongest macro-level performance when trained with Seq-CGAN augmentation. This outcome highlights the effectiveness of recurrent sequence modeling in capturing temporal intrusion patterns, particularly when combined with targeted synthetic data generation. More complex architectures, including attention-enhanced BiLSTM and Transformer-based models, do not consistently outperform simpler recurrent models in this imbalanced setting, underscoring that architectural complexity alone is insufficient without adequate data support.

Overall, the results confirm that informed sequence-level data augmentation is a powerful complement to loss reweighting and sampling strategies for intrusion detection. The proposed Seq-CGAN framework is model-agnostic, computationally practical, and can be integrated into existing IDS pipelines with minimal modification. By improving macro-level fairness and minority-class sensitivity while maintaining stable performance on dominant classes, Seq-CGAN-IDS offers a scalable and effective solution for real-world imbalanced cybersecurity datasets.

Future work will focus on adaptive augmentation strategies that dynamically target poorly performing classes during training, as well as cross-dataset generalization studies to assess the robustness and transferability of the proposed framework across different network intrusion benchmarks.

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